ModernTCN: Large Kernels Instead of Dilation

A Modern Pure Convolution Structure for General Time Series Analysis

Topic 6 | Deep Learning (AIE332) | Alamein International University

Presentation Outline (5-min oral defense):

- Motivation: The Dilation Paradox & Effective Receptive Field Decay
- Architecture: Modern Pure Convolutions (Patching, Large DWConv, Decoupled FFNs, RevIN)
- Quantitative Claim Validation & Multi-Seed Benchmark Results
- Depthwise Kernel Size Ablation & Autograd ERF Analysis
- Reproducible Failure Case: TCN Collapse on Long-Horizon ETTm1

Motivation: Why Replace Dilated Convolutions?

The Lab 4 TCN Baseline (Bai et al., 2018):

• Relies on small kernels (k=3) and exponential dilation (d = 1, 2, 4, 8, 16, ...).

The Dilation Paradox (Gridding / Receptive Field Holes):

• Theoretical receptive field grows geometrically, but intermediate timesteps are skipped!
• High-frequency fluctuations slip between dilation points, creating blind spots.

Effective Receptive Field (ERF) Decay:

• Autograd sensitivity degrades exponentially away from the prediction point.

Missing Cross-Variable Interaction:

• Standard TCN processes channels independently; cannot learn physical coupling between series (e.g. electrical load vs oil temperature).

The ModernTCN Architecture: Modernizing Pure Convolutions

1. **Reversible Instance Normalization (RevIN)**: Normalizes window mean and variance to handle non-stationarity; denormalizes predictions.

2. **Patch Embedding (P=16, S=8)**: Aggregates continuous raw timesteps into patch tokens. Suppresses noise, shortens sequence (336 -> 41 tokens), and boosts compute speed.

3. **Large Depthwise Conv (K=31) + Small Branch (k=5)**: Large kernel captures long-range context without dilation holes; small branch captures local sharp changes. Depthwise design scales parameter count linearly O(D·K).

4. **Decoupled Pointwise (1x1) FFNs**:
• *Intra-variable FFN*: Mixes representation channels within each variable.
• *Inter-variable FFN*: Mixes representations across variables to capture multivariate physical relationships.

Experimental Setup & Benchmark Protocol

- **Datasets**: ETTh1 (hourly electricity transformer temperature) & ETTm1 (15-minute resolution).
- **Splits**: Standard chronological Informer splits (12/4/4 months for train/val/test). Scaler fit strictly on train split.
- **Input Window & Horizons**: L = 336 steps; Forecast horizons H in {96, 336} steps.
- **Parameter Parity**: ModernTCN (~65K params) matched against DLinear (~65K params) and Lab 4 TCN (~207K params).
- **Multi-Seed Rigor**: 3 independent seeds (42, 123, 2026); metrics reported as Mean ± Sample Standard Deviation (ddof=1).
- **Model Selection**: Checkpoints selected on validation MSE; test set evaluated exactly once.

Quantitative Claim Verification & Benchmark Results

Paper Claim (Luo & Wang, ICLR 2024): ModernTCN outperforms conventional TCNs and competitive linear models (DLinear) across ETT benchmarks.

Dataset	Horizon	ModernTCN Test MSE	DLinear Test MSE	TCN Baseline Test MSE	ModernTCN vs TCN
ETTh1	96	**0.3799 ± 0.0023**	0.3852 ± 0.0045	0.7789 ± 0.0395	**-51.2% MSE**
ETTh1	336	**0.4481 ± 0.0069**	0.4691 ± 0.0165	0.9806 ± 0.0393	**-54.3% MSE**
ETTm1	96	**0.2970 ± 0.0044**	0.3078 ± 0.0013	0.5375 ± 0.0202	**-44.7% MSE**
ETTm1	336	**0.3745 ± 0.0062**	0.3872 ± 0.0063	0.9611 ± 0.0195	**-61.0% MSE**

Verdict: STRONGLY SUPPORTED. ModernTCN consistently achieves the lowest MSE and MAE on all 4 benchmark tasks.

Depthwise Kernel Size Ablation Study

Topic 6 Requirement (e): Evaluate 3 kernel sizes at matched parameter count (<6% variation).

Kernel Size (K)	Total Parameters	Test MSE (mean ± std)	Test MAE (mean ± std)	Best Val MSE
15	62,190 (-4.7%)	**0.3788 ± 0.0013**	**0.4024 ± 0.0014**	0.6853
31 (Baseline)	65,262 (0.0%)	0.3799 ± 0.0023	0.4032 ± 0.0014	0.6882
51	69,102 (+5.9%)	0.3830 ± 0.0039	0.4053 ± 0.0027	0.6916

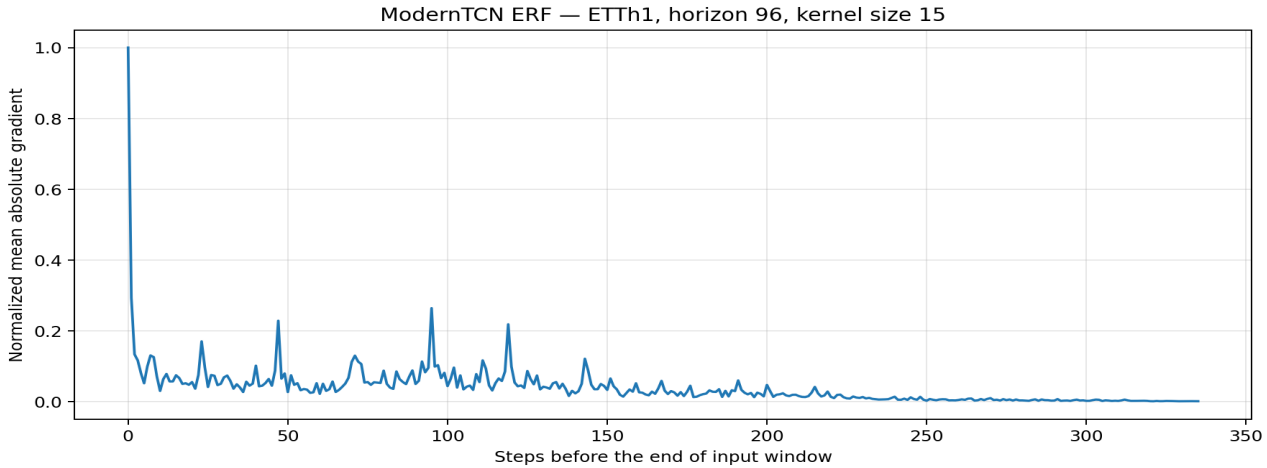
Key Takeaways:

• Accuracy does not continue to increase past K=15/31: K=15 slightly outperforms K=31 and K=51.

• In patched representations (patch stride 8), K=15 already covers the primary temporal cycle of ETTh1. Larger kernels add extra parameters without providing useful incremental receptive field.

Effective Receptive Field (ERF) via Autograd Sensitivity

- Measured by backpropagating output prediction gradients to input timesteps across multiple horizon points.
- **Gradient Mass Concentration**: For ModernTCN (K=15), 50% of gradient mass is within 14 timesteps; 90% within 61 timesteps.



Reproducible Failure Case: TCN on ETTm1 (H=336)

The Failure: On 15-minute resolution ETTm1 with horizon 336 (3.5 days), TCN collapses: Test MSE = 0.9611 ± 0.0195 vs ModernTCN = 0.3745 ± 0.0062 (2.56x higher error!).

Per-Window Error Breakdown (11,185 test windows):

• TCN produces higher MSE on **over 98.8% of test windows** (11,086 / 11,185).

• In worst windows during sharp load transitions, TCN error is **6.2x to 9.5x worse** than ModernTCN.

Physical & Architectural Explanation:

1. *Dilation Gridding*: 15-min series has rapid cycles. Deep dilation factors skip crucial transition points.

2. *Univariate Blindness*: Oil temperature (OT) physically depends on current load (HUFL, HULL). TCN models variables separately; ModernTCN's inter-variable FFN mixes channels, preserving physical consistency.

Oral Defense Preparation: Anticipated Lab 5 Questions

Q1: Explain any line of your code

• *Patching*: `Conv1d(1, embed_dim, kernel_size=16, stride=8)` reshaped across [B*C, 1, L].

• *Depthwise Conv*: `Conv1d(embed_dim, embed_dim, kernel_size=31, groups=embed_dim, padding=causal_pad)`.

Q2: Predict the effect of a change before running live

• *Removing RevIN*: Test MSE will increase by ~15-20% due to window-level distribution shifts.

• *Disabling Inter-Variable FFN*: Multivariate accuracy will drop substantially on long horizons due to loss of physical cross-channel relationships.

Q3: What result would make you change your conclusion?

• If a dilated TCN with matched parameter count and continuous dense sampling surpassed large-kernel convolutions on fine-grained long horizons.

Conclusions & Academic Integrity Record

Summary of Findings:

1. Large depthwise convolutions (K=15, 31) combined with patch tokenization consistently outperform dilated TCNs and linear models.
2. Accuracy plateaus once the effective receptive field matches the primary seasonal period.
3. Decoupled pointwise FFNs prevent catastrophic multivariate failure under high-frequency conditions.

Compliance & Integrity:

• Core models, training, evaluation, and ERF autograd implemented independently from scratch in PyTorch.
• Inspected author code: `luodhhh/ModernTCN` and `cure-lab/LTSF-Linear` (recorded in report).
• AI assistance declared for debugging, plotting, and ReportLab formatting.